Sec: SR.IIT_*COSC(MODEL-A) | SPCL GTM-14 | Date: 27-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**A**	2	**B**	3	**C**	4	**B**	5	**B**
6	**A**	7	**B**	8	**D**	9	**B**	10	**B**
11	**B**	12	**C**	13	**B**	14	**A**	15	**B**
16	**B**	17	**D**	18	**A**	19	**A**	20	**B**
21	**4**	22	**2**	23	**1**	24	**5**	25	**4**

CHEMISTRY

26	**A**	27	**B**	28	**B**	29	**B**	30	**B**
31	**D**	32	**A**	33	**D**	34	**D**	35	**D**
36	**B**	37	**C**	38	**B**	39	**A**	40	**A**
41	**D**	42	**D**	43	**D**	44	**C**	45	**D**
46	**5**	47	**2**	48	**0**	49	**2**	50	**0**

MATHEMATICS

51	**A**	52	**A**	53	**B**	54	**C**	55	**B**
56	**B**	57	**B**	58	**A**	59	**A**	60	**C**
61	**A**	62	**C**	63	**C**	64	**C**	65	**D**
66	**A**	67	**C**	68	**C**	69	**B**	70	**D**
71	**3 OR 4**	72	**0 OR 1**	73	**3**	74	**11**	75	**0**

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

1. **Question Explanation :**
We have to calculate the value of resistance R in the given potentiometer circuit.
Concept : This question is based on Application of Potentiometer.
Solution :

$$iR = \ell_1\phi$$

$$\left(\frac{E_s}{r+R}\right)R = \ell_1\phi$$

$$\left(\frac{E_1}{r_1+R}\right)R = \ell_1\phi \ldots(1)$$

$$\left(\frac{E_2}{r_2+R}\right)R = \ell_2\phi \ldots(2)$$

$$\left(\frac{E_1}{E_2}\right)\left(\frac{r_2+R}{r_1+R}\right) = \frac{\ell_1}{\ell_2}$$

$$\frac{10}{5}\left(\frac{2+R}{1+R}\right) = \frac{30}{10} = 3$$

$$2\left(\frac{2+R}{1+R}\right) = 3$$

$$4+2R = 3+3R \quad (2)$$

$$R = 1\Omega \quad (2)$$

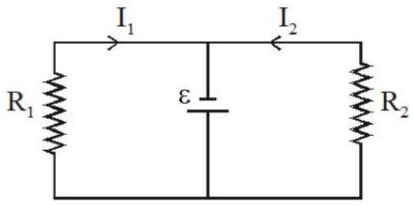
Hence,
Option (1) is correct : 1Ω

2. Rms current through resistor ' R_1 ' is

$$I_1(\text{rms}) = \frac{50}{\sqrt{2}\times 10\sqrt{2}} = \frac{5}{2}\ \text{A} \therefore \text{Average power developed in the resistor ' } R_1 \text{ ' is}$$

$$P_1 = I^2_{1(\text{rms})}R_1 = \left(\frac{5}{2}\right)^2 10 = \frac{250}{4} = 62.50\ \text{W}$$

3.



4. $(23.2 \pm 0.2)\,\text{mm}$

5. $\vec{E} = 10\hat{j}\cos(6x+8z)10\hat{j}\cos\left[\left(6\hat{i}+8\hat{k}\right)\cdot\left(x\hat{i}+z\hat{k}\right)\right]$

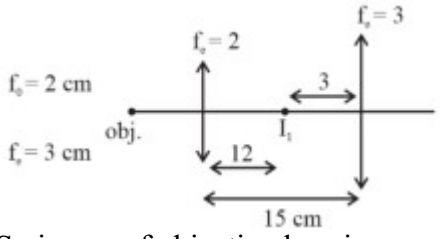
$\therefore \vec{k} = 6\hat{i}+8\hat{k}$: this is the direction of ' c' .

$\vec{c}\times\vec{E}$ is direction of $\vec{B}: \left(6\hat{i}+8\hat{k}\right)\times\hat{j} = 6\hat{k}-8\hat{i}$

$$\vec{B} = \left(\frac{6\hat{k}-8\hat{i}}{c}\right)\cos[6x+8z-10ct] \text{ as } B = \frac{E}{c}$$

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6. ∵ Image formed by eye-piece is at ∞, object is at focus of eye - piece.



So image of objective lens is
is at 12 cm
For objective lens

$$\frac{1}{v}-\frac{1}{4}=\frac{1}{f}$$

$$\frac{1}{12}-\frac{1}{4}=\frac{1}{2}$$

$$\frac{1}{4}=\frac{1}{12}-\frac{1}{2}=\frac{1-6}{12}$$

$$4=\frac{-12}{5}=-2.4$$

7. From $eV_0 = h\nu - \phi$ and $K_{max} = h\nu - \phi$. If ν increases keeping ϕ constant, then V_0 and K_{max} both increase.
If ϕ decreases keeping ν constant, then V_0 and K_{max} increase.
If I increases more photoelectrons would be liberated, hence saturation photocurrent increases.
If separation between cathode and anode is increased, then there is no effect on v_0, K_{max} or current.

8. Mosley law $\sqrt{\frac{e}{\lambda}}=\sqrt{\frac{3}{4}RC}(Z-1)$

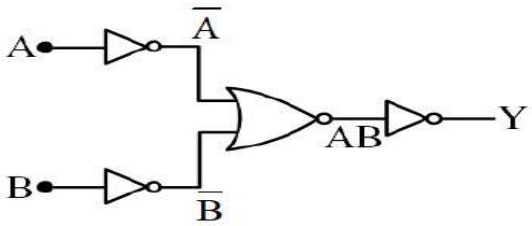
9. $\frac{(6A-1)}{(A+1)}$

10. $\frac{\mu_0 i}{4\pi R_1}\left(\frac{3\pi}{2}\right)+\frac{\mu_0 i}{4\pi R_2}\cdot\frac{\pi}{2}$

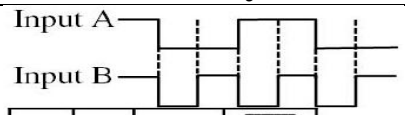
$$\frac{\mu_0 i}{8}\left[\frac{3}{R_1}+\frac{1}{R_2}\right]$$

11. $\phi = \vec{B}\cdot\vec{A}$
$= BA\cos\theta$
Most suitable ans is 2 [Otherwise ABCD]

12. Question Explanation: Determine the output waveform of a given logic circuit for specified input waveforms.
Concept: NAND gate operation and timing diagrams.
Solution:



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A	B	AB	$\overline{AB}$
1	1	1	0
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0
0	0	0	1
0	1	0	1

Option C matches with truth table

13. Question Explanation :
To find the final intensity of unpolarized light after passing through 3 different polarizing sheet.
Concept :
This question is based on Malus Law, Polarization of light.
Solution :
Intensity after unpolarized light passes through first polarizer.
$I' = \frac{I_0}{2}$ (Half of original intensity).
Angle between 1st and 2nd polarizer $= |30^\circ - 60^\circ| = 30^\circ = \theta_1$
Angle between 2nd and 3rd polarizer $= |60^\circ - 30^\circ| = 30^\circ$
$= \theta_2$
$I_{\text{final}} = I'\cos^2\theta_1, \cos^2\theta_2$
$= \frac{I_0}{2}\cos^2 30^\circ \cos^2 30^\circ = \frac{9I_0}{32}$
Hence,
Option (2) is correct

14. 5×10^{-6} m

15.
$$L = M^a\ T^b\left(M^1\ L^2\ T^{-2}\right)C$$
$$a + c = 0,\ b - 2c = 0$$
$$2c = 1 \Rightarrow c = \frac{1}{2}, a = -\frac{1}{2}$$
$$b = 1$$

16. Question Explanation:
Calculating time period for a V-shaped swing.
Concept:
Time Period of a Simple Pendulum
Solution:
The system acts as a simple pendulum. The effective length (l) is the vertical distance from the support to the mass M . Since the ropes and the support form an equilateral triangle of side L , we find the altitude (h).
Calculation:

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Effective length $h = L \cdot \sin\left(60^\circ\right) = \frac{\sqrt{3}\,L}{2}$.

Time Period $T = 2\pi\sqrt{\frac{h}{g}} = 2\pi\sqrt{\frac{\sqrt{3}\,L}{2\,g}}$.

Final Answer:

$2\pi\sqrt{\frac{\sqrt{3}\,L}{2\,g}}$

Option (2)

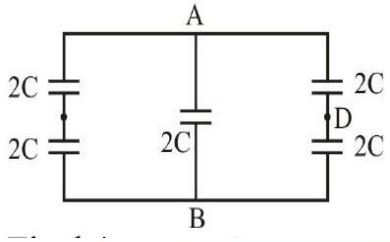
17. Question Explanation: The question asks to find the condition for individual rod lengths to remain unchanged when a composite rod system between fixed supports is heated.
Concept: Thermal stress in a constrained composite rod system
Solution:
Equating compressive forces, we get,
$A_1\left(\alpha,(\Delta\theta)\right)Y_1 = A_2\left(\alpha_2(\Delta\theta)\right)Y_2$
$\therefore \frac{A_1}{A_2} = \frac{\alpha_2 Y_2}{\alpha_1 Y_1}$
Final Answer: Option (4)

18. $2\frac{J}{m}$

19. $\frac{A}{T} =$ constant $\frac{A_1}{2} = \frac{A_2}{3} = \frac{A_3}{6}$

20. $\frac{\sqrt{3}\,g}{2}$

21. Explanation Question: We have to find the value of n where equivalent capacitance between A and B is nC
Concept: This question is based on wheat stone bridge
Solution: It is the balanced condition wheat stone bridge so the points C and D are at same potential



25. $\sqrt{2}\lambda_0\ T_1 = \lambda_0\ T_2$; $\frac{T_2}{T_1} = \sqrt{2}$; $\frac{A_2}{A_1} = \left(\frac{T_2}{T_1}\right)^4 = 4$

CHEMISTRY

27. According to $n+1$ rule electron will be filled in the following order of orbitals
$5s \rightarrow 4d \rightarrow 5p$

27. Question Explanation:
Match's the coloumn
Concept:
This question is based on perodic table
Solution:
Atomic number $= 8\left(1\,s^2\,2\,s^2\,2p^4\right) = (p\text{-block})$
$= 12\left(1\,s^2\,2\,s^2\,2p^6\,3\,s^2\right) \rightarrow (s-\text{ block})$

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$= 60\left(1\,s^2\,2\,s^2 2p^6 3\,s^2 3p^6 4\,s^2 3\,d^{10} 4p^6 5\,s^2\right.$

$\left.4\,d^{10} 5p^6 6\,s^2 4f^4\right) \rightarrow (f-\text{ block})$

$= 26\left(1\,s^2\,2\,s^2 2p^6 3\,s^2 3p^6 4\,s^2 3\,d^6\right) \rightarrow (d-\text{block})$

28. Fact based.

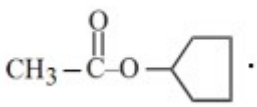
29. Explain the question : Sodium nitroprusside give a violet colour with which acidic radical in basic medium.
Concept : This question is based on test of sulphide.
Solution :
violet colour is with S^{-2} ion.
$Na_2\ S + Na_2\left[Fe(CN)_5(NO)\right] \rightarrow$
$Na_4\left[Fe(CN)_5(NOS)\right]$
(violet)

31. $H--C\equiv N\ H--C\equiv P$

32. Explanation Question: Find the difference in EAN for given complex
Concept: This question is based on EAN value
Solution:
For the complex $\left[Ti^{IV}(\sigma-C_5H_5)_2(\pi-C_5H_5)_2\right]$
$EAN = 22-4+4+12 = 34$
For the complex, $\left[Cr(C_6H_6)_2\right]$
$EAN = 24-0+12 = 36$
$\therefore$ Difference in EAN $= 36-34 = 2$

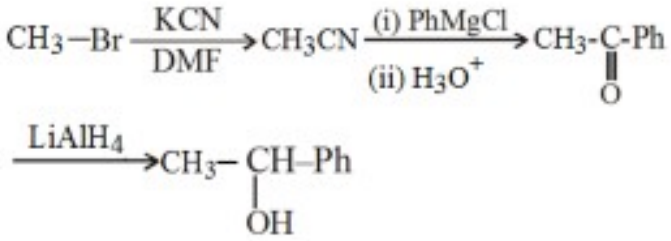
33. Wilkinson catalyst $-\left[RhCl(PPh_3)_3\right] = 16e^-$ cis-platin $-\left[PtCl_2(NH_3)_2\right] = 16e^-$
Brown ring complex $-\left[Fe(H_2O)_5\,NO\right]SO_4 = 19e^-$
Prussian blue $-Fe_4\left[Fe(CN)_6\right]_3 = 18e^-$

34. Question Explanation:
The question asks for the major product formed when an unsymmetrical ketone reacts with mCPBA under heat, based on group migration during oxidation.
Concept: Baeyer-Villiger Oxidation, Migratory Aptitude.
Solution:
m-CPBA converts ketones into esters via
Baeyer-Villiger oxidation.
In unsymmetrical ketones, the group with
higher migratory aptitude migrates.
Migratory order: tertiary alkyl > secondary
alkyl > primary alkyl > methyl.
Here, the cyclopentyl group has higher
migratory aptitude than CH_3, so it migrates. Oxygen is inserted between the carbonyl carbon and the cyclopentyl group, forming
the ester

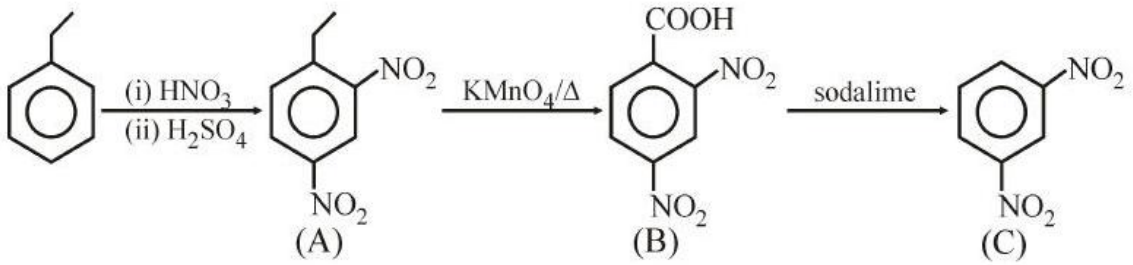


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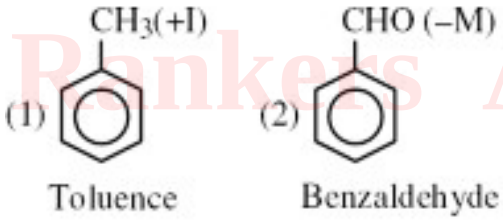
35.

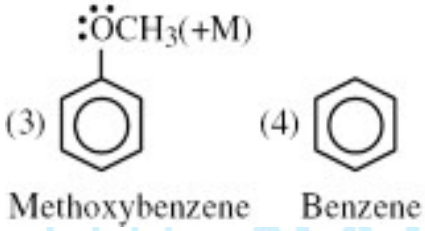


36.



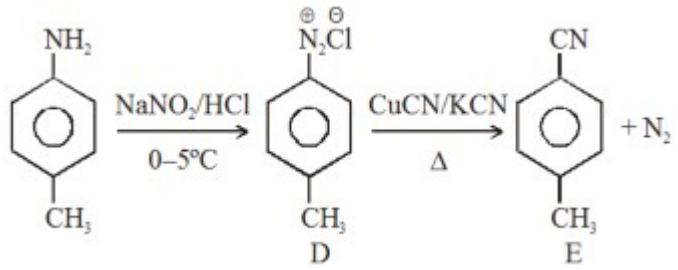
37. Explaining the Question :-
Question is asking about the most reactive aromatic compound towards electrophilic substitution reaction [ESR].
Concept:-
This question is based on rate of ESR $\propto +M / +I$ group attached directly to benzene
Solution :-
+M effect is more dominant over +I effect.





Hence option 3 is the most reactive towards ESR.

39.



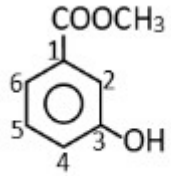
Formation of D is example of Diazotisation
Formation of E from D is example of Sandmayer's Reaction

40. A is unstable due to anti-aromaticity.

41. More the stable Anion better will be the leaving tendency.

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42. **Question Explanation:**
Determine the correct IUPAC name for a benzene ring substituted with an ester group ($-COOCH_3$) and a hydroxyl group (-OH).
Concept:
This question is based on rules for IUPAC nomenclature of organic compounds.
Solution:
The ester group has higher priority than the alcohol group, making the parent molecule methyl benzoate.
The carbon attached to the ester is $C-1$, and the ring is numbered to give the -OH group (hydroxy) the lowest possible number, which is 3 .
Combining these, the alkyl group from the ester is named first, followed by the substituent and the parent name.



Methyl 3-hydroxybenzoate

43. Question Explanation: Calculate the new equilibrium pressure after a volume change.
Concept: Chemical Equilibrium - Effect of volume change.
Solution: The equilibrium constant $\left(K_p\right)$ remains constant at a fixed temperature.
When the volume is doubled, the pressures initially halve, and the system shifts toward the side with more moles of gas (Le Chatelier's Principle) to reach a new equilibrium.
Reaction: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

Initial $K_p: \frac{\left(P_{NO_2}\right)^2}{P_{N_2O_4}} = \frac{0.4^2}{0.2} = 0.8$ atm

After doubling volume: Initial $P'_{NO_2} = 0.2$ and $P'_{N_2O_4} = 0.1$. Let x be the change in $P_{N_2O_4}$ to reach new equilibrium.
New Equilibrium pressures:
$\$P_\{\mathrm{N}\}_2\ \mathrm{O}\}_4\} = (0.1-x)\$$ and $\$P_\{\mathrm{NO}\}_2\} = (0.2+2x)\$$.
Solve for **x** :

$$0.8 = \frac{(0.2+2x)^2}{0.1-x} \rightarrow 0.08 - 0.8x = 0.04 + 0.8x + 4x^2 \rightarrow 4x^2 + 1.6x - 0.04 = 0 .$$

$x \approx 0.024$. New $P_{NO_2} = 0.2 + 2(0.024) \approx 0.248$ atm

44. $A(s) \rightleftharpoons B(g) + C(g) \#(i)$
At eq. 0.050.10.2
$2\,B(g) \rightleftharpoons D(g) + 2E(g) \ldots (ii) \#(ii)$
0.10.050.1$\#(ii)$

$$P_{Total} = \frac{(0.1+0.2+0.05+0.1)RT}{V} \#(ii)$$

$$= \frac{0.45 \times 0.08 \times 300}{1} = 10.8\ atm \#(ii)$$

$$K_{C_1} = 0.1 \times 0.2 = 0.02 \#(ii)$$

$$P_B = \frac{0.1}{0.45} \times 10.8 = 2.4\ atm \#(ii)$$

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45. Concept: Isotones are the atoms which have some no. of neutrones. In an atom ${}^{A}_{Z}X$, no. of neutrons can be calculated using relation $n = A - Z$

Solution :

A. ${}^{13}_{6}C$: no. of neutrons $= 13 - 6 = 7$

${}^{14}_{7}N$: no. of neutrons $= 14 - 7 = 7$

These are isotones.

B. ${}^{31}_{15}P$: neutrons $= 31 - 1516$

${}^{32}_{16}S$: neutrons $= 32 - 1516$

These are isotones

C. ${}^{30}_{14}Si \diagdown \&{}^{31}_{15}P$

${}^{30}_{14}Si$: neutrons $= 16$

${}^{31}_{15}P$: neutrons $= 16$

These are isotones.

D. ${}^{11}_{5}B$: neutrons $= 6$

${}^{13}_{6}C$: neutrons $= 7$

These are not isotones, Hence correct answer is (D) ${}^{30}_{14}Si \& {}^{31}_{15}P$

46. $H_2O(s) \xrightarrow[\Delta s_1]{} H_2O(s) \rightleftharpoons_{\Delta s_2} H_2O(\ell) \xrightarrow[\Delta s_3]{} H_2O(\ell) - 10^{\circ}C\,0^{\circ}C\,0^{\circ}\,10^{\circ}C$

$$As = 1\times 10\ln\left(\frac{273}{263}\right) + \frac{1092}{273} + 1\times 18\ln\left(\frac{283}{273}\right)$$

$$= 10\times 0.037 + 4 + 18\times 0.035$$

$$= 0.37 + 4 + 0.63 = 5$$

47. $3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$

$$\text{Mole of } Cu(NO_3)_2 = \frac{18.75}{187.5} = 0.1$$

$$\Rightarrow \text{Mole of NO} = \left[\ \text{Mole of } Cu(NO_3)_2\right] \times \frac{2}{3} = \frac{0.2}{3} \times 30 = 2\ g.$$

48. $Ag + \rightarrow \underset{\substack{0.1M \\ Ag^+ \\ ande}}{Ag^+} + Ag$

$$K_f = \frac{\left[Aq^+\right](NH_3)^2}{\left(Ag(NH_3)_2^+\right)} \Rightarrow 10^8 = \frac{\left[Ag^+\right]\text{anode} \times 1^2}{(0.01)}$$

$$[\ Ag^+\ ]\ \text{anode} = 10^6$$

$$E = E^{\circ} - \frac{0.06}{1}\log\frac{\left[Ag^+\right]\text{cathode}}{\left[Ag^+\right]\text{anode}}$$

$$E = 0 - 0.06\log\frac{10^{-1}}{10^6}$$

$$= 0.06\times 7 = 0.42 \text{ volt}.$$

49. Question Explanation: Determination of reaction order from the half-life graph.

Concept: Half-life of nth Order Reaction.

Solution: The relationship between half-life ($t_{1/2}$) and initial concentration (a_0) for an nth order reaction is $t_{1/2} \propto a_0{}^{1-n}$.

Taking logs: $\log t_{1/2} = (1-n)\log a_0 + \log C$.

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The slope of the graph is $(1-n)$. So, slope (m)

$= \tan(135^\circ) = -1$.

Since $m = 1-n$, then $-1 = 1-n$.

Thus, $n = 1+1 = 2$.

Final Answer: 2

50. $k = \frac{1}{t}\ln\frac{r_\infty - r_o}{r_\infty - r_t} = \frac{1}{10}\ln\frac{-10-20}{-10+2.5} = \frac{1}{10}\ln 4$

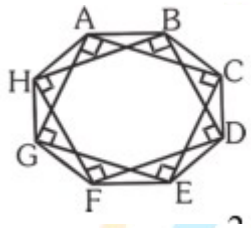
MATHS

51. Question Explanation: Probability that the quadrilateral formed by four randomly chosen vertices of a regular octagon is a square.

Concept: Combinations and Geometric Constraints.

Solution:

ACEG and BDFH is



a square $P = \frac{2}{{}^8C_4} = \frac{1}{35}$

52. Let m be the slope of line which passing, through (h, k)

$\therefore$ equation of line $y-k = m(x-h)$

eq. of line perpendicular to (1) & passing through $(0,0)$ is

$y-0 = -\frac{1}{m}(x-0)\#(2)$

from (1) & (2) eliminate m , we get

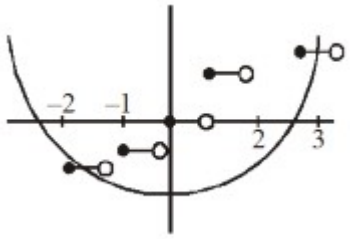
$x^2+y^2-hx-ky=0$, which is the required locus of the point

53. $-1 \le [x] - x^2 + 4 \ge 2$

$-5 \le [x] - x^2 \le -2$

(i) $-5 \le [x] - x^2 \Rightarrow x^2 - 5 \le [x]$

Intersection for both the curves above x axis lies between $(2,3)$



$x^2 - 5 = 2 \Rightarrow x = \sqrt{7}$

below x axis between $(-2,-1)$

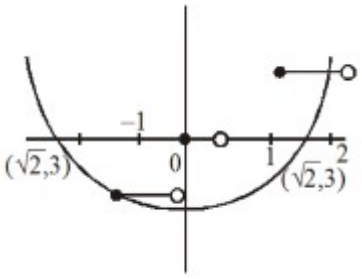
$x^2 - 5 = -2 \Rightarrow x = -\sqrt{3}$

$x = -\sqrt{3}$

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(ii) $[x]-x^2 \le -2 \Rightarrow x^2-2 \ge [x]$

above x -axis between $(1,2)$



$x^2-2=1 \Rightarrow x=\sqrt{3}$

below axis it is at $x=-1$

The interval is $(-\infty,-1] \cup (\sqrt{3},\infty)$

Intersection of case (i) & (ii) gives option (B)

54. $m \ge 12$

$0 \le n \le 50$

$\ell \ge 19$

55. $\hat{a},\hat{b}$ and $\hat{c}$ are unit vectors. Now

$x=|\hat{a}-\hat{b}|^2+|\hat{b}-\hat{c}|^2+|\hat{c}-\hat{a}|^2$

$=2\left(\hat{a}\cdot\hat{a}+\hat{b}\cdot\hat{b}+\hat{c}\cdot\hat{c}\right)-2\hat{a}\cdot\hat{b}-2\hat{b}\cdot\hat{c}-2\hat{c}\cdot\hat{a}$

$=6-2\left(\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}+\hat{c}\cdot\hat{a}\right)\#(i)$

Also, $|\hat{a}+\hat{b}+\hat{c}| \ge 0$

or $\hat{a}\cdot\hat{a}+\hat{b}\cdot\hat{b}+\hat{c}\cdot\hat{c}+2\left(\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}+\hat{c}\cdot\hat{a}\right) \ge 0$

or $3+2\left(\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}+\hat{c}\cdot\hat{a}\right) \ge 0$

or $2\left(\hat{a}\cdot\hat{b}+\hat{b}\cdot\hat{c}+\hat{c}\cdot\hat{a}\right) \ge -3$

from (1) $x \ge 9$

56. We know that $(ABC)^{-1}=C^{-1}B^{-1}A^{-1}$

$\left(A(A+B)^{-1}B\right)^{-1}=\left(B^{-1}\right)\left((A+B)^{-1}\right)^{-1}A^{-1}=B^{-1}$

$(A+B)A^{-1}$

$=\left(B^{-1}A+B^{-1}B\right)A^{-1}=\left(B^{-1}A+I\right)A^{-1}=B^{-1}\left(AA^{-1}\right)+I^{-1}=B^{-1}I+A^{-1}=B^{-1}+A^{-1}$

57. $\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$

$=\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\frac{2n(n+1)}{2}\right)\right)$

$=\cot\left(\sum_{n=1}^{23}\tan^{-1}\frac{1}{1+n(n+1)}\right)$

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$$= \cot\left(\sum_{n=1}^{23} \tan^{-1}\frac{(n+1)-n}{1+n(n+1)}\right)$$

$$= \cot\left(\sum_{n=1}^{23}\left(\tan^{-1}(n+1)-\tan^{-1}n\right)\right)$$

$$= \cot\left(\tan^{-1}24-\tan^{-1}1\right)$$

$$= \cot\left(\tan^{-1}\frac{23}{25}\right) = \cot\left(\cot^{-1}\left(\frac{25}{23}\right)\right) = \frac{25}{23}$$

58. Let $T_r = r(n-r)$

$$T_r = nr - r^2$$

$$\Rightarrow S_n = \sum_{r=1}^{n} T_r = \sum_{r=1}^{n}\left(nr-r^2\right)$$

$$S_n = \frac{n\cdot(n)(n+1)}{2} - \frac{n(n+1)(2n+1)}{6} \Rightarrow$$

$$S_n = \frac{n(n-1)(n+1)}{6}$$

Now $\sum_{n=4}^{\infty}\left(\frac{2\,S_n}{n!} - \frac{1}{(n-2)!}\right) =$

$$\sum_{n=4}^{\infty}\left(2.\frac{n(n-1)(n+1)}{6.n(n-1)(n-2)!} - \frac{1}{(n-2)!}\right)$$

$$= \sum_{n=4}^{\infty}\left(\frac{1}{3}\left(\frac{n-2+3}{(n-2)!}\right) - \frac{1}{(n-2)!}\right)$$

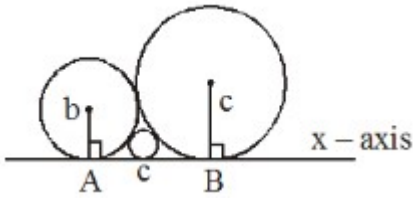
$$= \sum_{n=4}^{\infty}\frac{1}{3}\cdot\frac{1}{(n-3)!} = \frac{1}{3}(e-1)$$

$$\left[e = \frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\ldots\right]$$

59. $AB = AC + CB$

$$\sqrt{(b+c)^2-(b-c)^2}$$

$$= \sqrt{(b+a)^2-(b-a)^2} + \sqrt{(a+c)^2-(a-c)^2}$$



$$\sqrt{bc} = \sqrt{ab} + \sqrt{ac}$$

$$\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{c}} + \frac{1}{\sqrt{b}}$$

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60. $\beta = \lim_{x\to 0} \dfrac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$

$\beta = \lim_{x\to 0} \dfrac{1 + \alpha x - \left[1 + 3x + \dfrac{9x^2}{2!} + \ldots\ldots\right]}{(\alpha x)\dfrac{\left(e^{3x-1}\right)}{3x}3x}$

$\beta = \lim_{x\to 0} \dfrac{(\alpha x - 3x) - \dfrac{9x^2}{2!} - \ldots\ldots}{3\alpha x^2}$

For existence of limit $\alpha - 3 = 0$

$\alpha = 3$

Limit $\beta = \dfrac{-3}{2\alpha}$

$\beta = -\dfrac{1}{2}$

Now,

$\alpha + \beta = \dfrac{5}{2}$

61. $(2021)^{2022} + (2022)^{2021}$

$= (2023 - 2)^{2022} + (2023 - 1)^{2021}$

$= 7n_1 + 2^{2022} + 7n_2 - 1$

$= 7(n_1 + n_2) + 8^{674} - 1$

$= 7(n_1 + n_2) + (7 + 1)^{674} - 1$

$= 7(n_1 + n_2) + 7n_3 + 1 - 1$

$= 7(n_1 + n_2 + n_3)$

$\therefore$ Given number is divisible by 7 hence remainder is zero

62. $f(x) = \int_0^1 (5 + (1 - t))\,dt + \int_1^x (5 + (t - 1))\,dt$

$= 6 - \dfrac{1}{2} + \left(4t + \dfrac{t^2}{2}\right)\Big|_1^x$

$= \dfrac{11}{2} + 4x + \dfrac{x^2}{2} - 4 - \dfrac{1}{2}$

$= \dfrac{x^2}{2} + 4x + 1$

$f(2^+) = 2 + 8 + 1 = 11$

$f(2) = f(2^-) = 5 \times 2 + 1 = 11$

$\Rightarrow$ continuous at $x = 2$

Clearly differentiable at $x = 1$

$Lf'(2) = 5$

$Rf'(2) = 6$

$\Rightarrow$ not differentiable at $x = 2$

63. **Explain Question:**

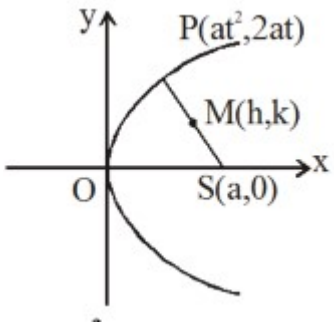
Finding the directrix of the focus of the mid point between focus and a moving point.

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Concept:

Focus of mid-points on a parabola.

Solution:



$h=\frac{at^2+a}{2}, k=\frac{2at+0}{2}$

$\Rightarrow t^2=\frac{2h-a}{a}$ and $t=\frac{k}{a}$

$\Rightarrow \frac{k^2}{a^2}=\frac{2h-a}{a}$

$\Rightarrow$ Locus of (h,k) is $y^2=a(2x-a)$

$\Rightarrow y^2=2a\left(x-\frac{a}{2}\right)$

Its directrix is $x-\frac{a}{2}=-\frac{a}{2}\Rightarrow x=0$

Final Answer: Option (3)

64. $f'(x)=(x-3)^{n_1-1}(x-5)^{n_2-1}(n_1+n_2)\left(x-\frac{5n_1+3n_2}{n_1+n_2}\right)$

Option (3) is incorrect since

for $n_1=3, n_2=5$

$f'(x)=8(x-3)^2(x-5)^4\left(x-\frac{30}{8}\right)$

minima at $x=\frac{30}{8}$

65. $x^2+9y^2-4x+3=0$

$(x^2-4x)+(9y^2)+3=0$

$(x^2-4x+4)+(9y^2)+3-4=0$

$(x-2)^2+(3y)^2=1$

$\frac{(x-2)^2}{(1)^2}+\frac{y^2}{\left(\frac{1}{3}\right)^2}=1$ (equation of an ellipse).

As it is equation of an ellipse, x & y can vary inside the ellipse.

So, $x-2\in[-1,1]$ and $y\in\left[-\frac{1}{3},\frac{1}{3}\right]$

$x\in[1,3]\, y\in\left[-\frac{1}{3},\frac{1}{3}\right]$

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66. $\because \int \frac{x^3}{\left(2x^2+1\right)^3}\cdot dx$

$= \int \frac{x^3}{x^6\left(2+x^{-2}\right)^3}\cdot dx$

$= \int \frac{x^{-3}\cdot dx}{\left(2+x^{-2}\right)^3}$ { Put $2+x^{-2}=t \Rightarrow$

$x^{-3}.dx = \frac{dt}{-2}$ }

$= -\frac{1}{2}\int \frac{dt}{t^3}$

$= -\frac{1}{2}\left(-\frac{1}{2}\cdot\frac{1}{t^2}\right)+c$

$= \frac{1}{4}\left(2+\frac{1}{x^2}\right)^{-2}+c$

67. For the line to be co-planar

$$\begin{vmatrix} 2 & 0 & k \\ 1 & k & -2 \\ 4 & 1 & -1-k \end{vmatrix} = 0$$

$\Rightarrow 2\left(-k-k^2+2\right)+k\left(1-4k\right)=0$

$-6k^2-k+4=0$

$6k^2+k-4=0$

This equation has two distinct roots So two such planes exist.

68. Let $a+b=x$ then $ab=\frac{x^2-7}{2}$

and $a^3+b^3=10 \Rightarrow (a+b)^3-3ab\left(a+b\right)=10$

$\Rightarrow x^3-3\left(\frac{x^2-7}{2}\right)x=10 \Rightarrow x^3-21x+20=0$

$\Rightarrow x=1,4,-5$

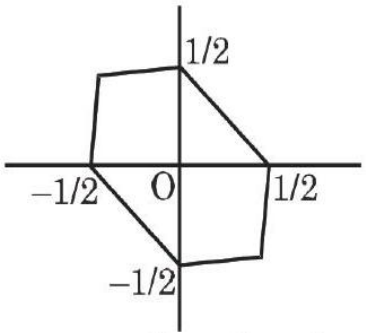
69. **Question Explanation:**

Finding the area of a region defined by an inequality with absolute values.

Concept:

This question is based on area of a region defined by absolute values.

Solution:



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$$A = 2\times\frac{1}{2}\times\frac{1}{2}+\frac{1}{2}\times\frac{1}{2}=\frac{3}{4}$$

Final Answer:

The correct option is (2)

70. Question Explanation: Calculate the unknown variance of one set when two sets are combined.
Concept: Combined Variance of two sets. Solution:

$$\text{Combine var.} = \frac{n_1\sigma^2+n_2\sigma^2}{n_1+n_2}+\frac{n_1n_2\left(m_1-m_2\right)^2}{\left(n_1+n_2\right)^2}$$

$$13=\frac{15.14+15.\sigma^2}{30}+\frac{15.15(12-14)^2}{30\times30}$$

$$13=\frac{14+\sigma^2}{2}+\frac{4}{4}$$

$$\sigma^2=10$$

71. Let $x=2K_1+1, y=2K_2+1, z=2K_3+1$

$$\therefore K_1+K_2+K_3=25\#(1)$$

Number of Non negative integeral solutions $={}^{27}C_2,\ N=351$

72. Using king in ' m '

$$m=\int_{-1}^{2}(1-x)\sin\left((1-x)x\right)dx$$

$$=\int_{-1}^{2}\sin\left((1-x)x\right)dx-\int_{-1}^{2}x\sin\left((1-x)x\right)dx$$

$$m=n-m\Rightarrow 2m=n \text{ or } \frac{m}{n}=\frac{1}{2}$$

73. $\sin^6\theta+\cos^6\theta=\frac{1}{2}$

$$\Rightarrow 1-3\sin^2\theta\cos^2\theta=\frac{1}{2}$$

$$\Rightarrow \sin^2\theta\cos^2\theta=\frac{1}{6}$$

$$\sin^4\theta+\cos^4\theta=\lambda_2$$

$$\Rightarrow \lambda_2=1-2\sin^2\theta\cos^2\theta=\frac{2}{3}$$

$$\lambda_3=\sin^4\theta\cos^4\theta=\frac{1}{36}$$

$$\frac{\lambda_1}{10}=\sin^6\theta\cos^6\theta=\frac{1}{216}\Rightarrow\lambda_1=\frac{5}{108}$$

$$\Rightarrow 36\lambda_1+\lambda_2+24\lambda_3=\frac{5}{3}+\frac{2}{3}+\frac{2}{3}=3$$

74. Put $z=\omega$ or $z=w^2$.

75. $C_2\to C_2-\left(\frac{C_1+C_3}{2}\right)$

$$=\begin{vmatrix}k+2 & 0 & k+4\\ k+3 & 0 & k+5\\ k+2p & 0 & k+2r\end{vmatrix};\ \text{use } p+r=2q$$

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